

Manoj Kumar Eede

manojkumareede@gmail.com | +91 8688640213 | manoj4689.github.io/personal-portfolio
github.com/Manoj4689 | linkedin.com/in/manojkumareede

Technical Skills

Languages: Java, Python, C++, JavaScript, Swift, SQL

Libraries & Frameworks: Spring Boot, Flask, React, LangGraph, LangSmith, PyTorch, TensorFlow, Scikit-learn

Databases & Architecture: MySQL, PostgreSQL, Redis, Kafka, Microservices, REST APIs, Distributed Systems, RAG Pipelines

Cloud & DevOps: AWS, Azure, GCP, Kubernetes, Docker, Argo CD, GitHub Actions, Splunk, Prometheus, CI/CD, MLOps

Work Experience

Adobe

Noida, India

Member of Technical Staff II

Jan 2026 – Present

- Orchestrated external rollout of PDF conversion backend microservices as a Java REST API, enabling enterprise LLM training; engineered multi-language SDKs (Java, Python, .NET, Node.js), projected to generate **\$1M ARR** in the initial phase.
- Architected a scalable PDF digitization backend service converting handwritten documents into structured digital formats using computer vision and multimodal LLMs; powering handwriting capabilities across Adobe AI applications (Study Spaces, Assistant).
- Designed and deployed a real-time document classification gateway (CNNs, Vision Transformers, transfer learning) that categorizes scanned pages and routes them to appropriate downstream document AI micro-services, processing **2M+** pages monthly.

Member of Technical Staff

Sept 2024 – Jan 2026

- Architected 0-to-1 PDF-to-Markdown and Markdown-to-PDF pipelines across distributed systems (Python and Java microservices); integrated into **ChatGPT** and **Copilot** via Acrobat SDK (MCP tools) for AI-assisted document workflows at scale.
- Owned Adobe's core Java-based DocGen API, reducing error rates to **sub-0.01%** across **300K+** monthly enterprise transactions.
- Built a GenAI-powered financial document assistant as a microservice with an end-to-end **RAG** pipeline (ingestion, chunking, vector retrieval, guardrails) serving extraction, multi-hop reasoning, and insight generation over enterprise financial data.
- Tech Stack:** Java, Spring Boot, Python, React, Redis, PostgreSQL, LangChain, FAISS, Kubernetes, Docker, Argo CD, Splunk

Asper.ai

Bangalore, India

Data Scientist

June 2024 – Sept 2024

- Led end-to-end client engagement for a **\$3M** contract with a leading US-based CPG company (1,000+ SKUs), spanning EDA, demand forecasting algorithm design, and post-production RCA; improved demand planning accuracy by over **10%**.
- Developed and benchmarked time-series forecasting models with automated retraining pipelines for production deployment.
- Built ML serving REST APIs for real-time and batch forecast delivery with model drift monitoring and performance tracking.
- Tech Stack:** Python, FastAPI, TensorFlow, Pandas, Scikit-learn, Azure, Kubernetes, MLOps

Adobe Research Lab

Bangalore, India

Research Intern

May 2023 – July 2023

- Built an end-to-end Automated Commenting Pipeline as a Flask microservice using RAG with hybrid embeddings (semantic + lexical) and prompt engineering, achieving **15%** higher accuracy on NLPeer dataset. Shipped into Adobe's GenAI framework.
- Fine-tuned Flan-T5 and BERT for Grounded Text Generation with custom evaluation benchmarks to optimize reviewer efficiency.
- Tech Stack:** Python, Flask, React, PyTorch, Hugging Face Transformers

Open Source Contributions

Docx4j – Corrupted Documents Fix (GitHub)

Docx4j is a widely used open-source Java library for Microsoft Office document processing across thousands of enterprise applications. Identified and helped resolve a critical DOCX corruption bug; additionally coordinated with the Apache FOP team to streamline an interdependent release. Fix merged into Docx4j's official release, directly improving document reliability for downstream users.

Education

Indian Institute of Technology, Roorkee

Roorkee, India

Integrated Masters in Mathematics and Computing (Applied Mathematics)

July 2019 – May 2024

CGPA: 8.78 / 10 **Coursework:** OOP, DBMS, DSA, OS, Discrete Mathematics, Graph Theory, Digital Image Processing

Projects

FinGen – Agentic-AI Powered Investment Assistant

2025 – 2026

Built a full-stack agentic AI platform for Indian stock market insights, sentiment analysis, and portfolio tracking. Deployed on Hugging Face Spaces with load balancing, caching, and CI/CD. *Stack: PyTorch, SmolAgents, Transformers, Supabase, Streamlit.*

CommerceHub – Cloud-Native E-Commerce Platform (GitHub)

2024

Built a cloud-native e-commerce platform with JWT auth, React frontend, and Spring Boot backend with Redis caching, PostgreSQL, and unit testing. *Stack: Java, Spring Boot, React, Redis, PostgreSQL, Docker, JUnit.*

Publications

Demand Forecasting Using Statistical Machine Learning Models.

[Springer Nature, forthcoming '26]

Madhu Jain, **Manoj Kumar Eede**, Nidhi Sharma.

In *Advanced Machine Learning and Evolutionary Computing Algorithms for Financial Decision-Making*.

Awards and Achievements

- SPOT Award (2025):** Recognized by Adobe for outstanding contributions to AI/ML initiatives.
- INSPIRE Scholarship, DST MHRD (2019–2024):** Awarded to top 1% students nationwide in pure sciences.
- Competitive Programming:** Codeforces – **1664** | **Expert** (Handle: codehacker4689), LeetCode – **2047** (Handle: 19312009).

Certifications

AI Agents (Hugging Face) · Model Context Protocol (Hugging Face) · Time Series Analysis, Forecasting & ML (Udemy) · Deep Learning Specialization (Udemy) · ML Regression Master Class in Python (Udemy)